

Method	E	ppl	% Knowledge Acquisition Performance (Δ)			Entity Learning			% Out of Domain Performance (Δ)		
			Avg.	High-Freq.	Low-Freq.	#Acq	#Ret	#Forg.	Avg.	MMLU	Social-IQA
Base	0	–	41	49	33	17	–	–	64	50	78
LoRA	1	8.54	46 ($\uparrow$ 5)	55 ($\uparrow$ 6)	37 ($\uparrow$ 4)	12	15	2	56 ($\downarrow$ 8)	45 ($\downarrow$ 5)	67 ($\downarrow$ 11)
	2	8.28	45 ($\uparrow$ 4)	55 ($\uparrow$ 6)	35 ($\uparrow$ 2)	3	24	3	54 ($\downarrow$ 10)	45 ($\downarrow$ 5)	63 ($\downarrow$ 15)
	3	8.15	44 ($\uparrow$ 3)	53 ($\uparrow$ 4)	34 ($\uparrow$ 1)	4	23	4	57 ($\downarrow$ 7)	46 ($\downarrow$ 4)	67 ($\downarrow$ 11)
	4	8.05	41 (= 0)	50 ($\uparrow$ 1)	32 ($\downarrow$ 1)	1	24	3	56 ($\downarrow$ 8)	45 ($\downarrow$ 5)	68 ($\downarrow$ 10)
	5	7.96	43 ($\uparrow$ 2)	51 ($\uparrow$ 2)	34 ($\uparrow$ 1)	4	23	2	55 ($\downarrow$ 9)	45 ($\downarrow$ 5)	66 ($\downarrow$ 12)
	6	7.89	39 ($\downarrow$ 2)	49 (= 0)	30 ($\downarrow$ 3)	1	20	7	56 ($\downarrow$ 8)	44 ($\downarrow$ 6)	68 ($\downarrow$ 10)
	7	7.83	40 ($\downarrow$ 1)	49 (= 0)	31 ($\downarrow$ 2)	2	19	2	56 ($\downarrow$ 8)	45 ($\downarrow$ 5)	66 ($\downarrow$ 12)
	8	7.76	38 ($\downarrow$ 3)	48 ($\downarrow$ 1)	29 ($\downarrow$ 4)	0	17	4	55 ($\downarrow$ 9)	44 ($\downarrow$ 6)	66 ($\downarrow$ 12)
	9	7.70	37 ($\downarrow$ 4)	45 ($\downarrow$ 4)	29 ($\downarrow$ 4)	2	14	3	58 ($\downarrow$ 6)	44 ($\downarrow$ 6)	72 ($\downarrow$ 6)
	10	7.65	35 ($\downarrow$ 6)	43 ($\downarrow$ 6)	27 ($\downarrow$ 6)	4	9	7	56 ($\downarrow$ 8)	43 ($\downarrow$ 7)	69 ($\downarrow$ 9)
Step-wise Regular.	1	8.74	47 ($\uparrow$ 6)	56 ($\uparrow$ 7)	39 ($\uparrow$ 6)	12	16	1	61 ($\downarrow$ 3)	49 ($\downarrow$ 1)	73 ($\downarrow$ 5)
	2	8.36	49 ($\uparrow$ 8)	56 ($\uparrow$ 7)	42 ($\uparrow$ 9)	5	28	0	60 ($\downarrow$ 4)	49 ($\downarrow$ 1)	71 ($\downarrow$ 7)
	3	8.19	48 ($\uparrow$ 7)	56 ($\uparrow$ 7)	40 ($\uparrow$ 7)	0	28	5	59 ($\downarrow$ 5)	48 ($\downarrow$ 2)	69 ($\downarrow$ 9)
	4	8.08	48 ($\uparrow$ 7)	55 ($\uparrow$ 6)	41 ($\uparrow$ 8)	3	28	0	58 ($\downarrow$ 6)	48 ($\downarrow$ 2)	68 ($\downarrow$ 10)
	5	7.99	45 ($\uparrow$ 4)	54 ($\uparrow$ 5)	36 ($\uparrow$ 3)	0	28	3	58 ($\downarrow$ 6)	47 ($\downarrow$ 3)	69 ($\downarrow$ 9)
	6	7.92	43 ($\uparrow$ 2)	52 ($\uparrow$ 3)	35 ($\uparrow$ 2)	2	26	2	60 ($\downarrow$ 4)	48 ($\downarrow$ 2)	72 ($\downarrow$ 6)
	7	7.85	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	34 ($\uparrow$ 1)	1	23	5	58 ($\downarrow$ 6)	47 ($\downarrow$ 3)	69 ($\downarrow$ 9)
	8	7.79	41 (= 0)	49 (= 0)	34 ($\uparrow$ 1)	3	19	5	57 ($\downarrow$ 7)	46 ($\downarrow$ 4)	67 ($\downarrow$ 11)
	9	7.73	41 (= 0)	48 ($\downarrow$ 1)	35 ($\uparrow$ 2)	4	17	5	56 ($\downarrow$ 8)	45 ($\downarrow$ 5)	67 ($\downarrow$ 11)
	10	7.68	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	33 (= 0)	3	16	5	56 ($\downarrow$ 8)	45 ($\downarrow$ 5)	68 ($\downarrow$ 10)
Pre-training Regular.	1	9.34	41 (= 0)	48 ($\downarrow$ 1)	34 ($\uparrow$ 1)	3	15	2	64 (= 0)	50 (= 0)	79 ($\uparrow$ 1)
	2	9.26	41 (= 0)	50 ($\uparrow$ 1)	32 ($\downarrow$ 1)	4	16	2	63 ($\downarrow$ 1)	49 ($\downarrow$ 1)	77 ($\downarrow$ 1)
	3	9.24	42 ($\uparrow$ 1)	49 (= 0)	34 ($\uparrow$ 1)	2	18	2	63 ($\downarrow$ 1)	49 ($\downarrow$ 1)	76 ($\downarrow$ 2)
	4	9.23	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	33 (= 0)	2	16	4	63 ($\downarrow$ 1)	50 (= 0)	76 ($\downarrow$ 2)
	5	9.21	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	33 (= 0)	2	16	2	65 ($\uparrow$ 1)	50 (= 0)	80 ($\uparrow$ 2)
	6	9.20	43 ($\uparrow$ 2)	51 ($\uparrow$ 2)	35 ($\uparrow$ 2)	5	14	4	63 ($\downarrow$ 1)	50 (= 0)	77 ($\downarrow$ 1)
	7	9.19	43 ($\uparrow$ 2)	52 ($\uparrow$ 3)	34 ($\uparrow$ 1)	4	16	3	63 ($\downarrow$ 1)	50 (= 0)	77 ($\downarrow$ 1)
	8	9.19	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	33 (= 0)	2	18	2	63 ($\downarrow$ 1)	50 (= 0)	77 ($\downarrow$ 1)
	9	9.18	43 ($\uparrow$ 2)	50 ($\uparrow$ 1)	35 ($\uparrow$ 2)	1	17	3	64 (= 0)	50 (= 0)	78 (= 0)
	10	9.16	43 ($\uparrow$ 2)	51 ($\uparrow$ 2)	35 ($\uparrow$ 2)	4	16	2	62 ($\downarrow$ 2)	50 (= 0)	75 ($\downarrow$ 3)
Structured Annotation	1	4.89	44 ($\uparrow$ 3)	52 ($\uparrow$ 3)	37 ($\uparrow$ 4)	11	14	3	57 ($\downarrow$ 7)	44 ($\downarrow$ 6)	71 ($\downarrow$ 7)
	2	4.74	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	34 ($\uparrow$ 1)	5	22	3	53 ($\downarrow$ 11)	43 ($\downarrow$ 7)	63 ($\downarrow$ 15)
	3	4.68	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	34 ($\uparrow$ 1)	1	22	5	55 ($\downarrow$ 9)	44 ($\downarrow$ 6)	67 ($\downarrow$ 11)
	4	4.63	43 ($\uparrow$ 2)	51 ($\uparrow$ 2)	36 ($\uparrow$ 3)	4	19	4	55 ($\downarrow$ 9)	44 ($\downarrow$ 6)	67 ($\downarrow$ 11)
	5	4.59	43 ($\uparrow$ 2)	52 ($\uparrow$ 3)	34 ($\uparrow$ 1)	3	21	2	54 ($\downarrow$ 10)	43 ($\downarrow$ 7)	65 ($\downarrow$ 13)
	6	4.56	43 ($\uparrow$ 2)	51 ($\uparrow$ 2)	35 ($\uparrow$ 2)	2	22	2	56 ($\downarrow$ 8)	44 ($\downarrow$ 6)	67 ($\downarrow$ 11)
	7	4.53	42 ($\uparrow$ 1)	50 ($\uparrow$ 1)	34 ($\uparrow$ 1)	3	17	7	54 ($\downarrow$ 10)	45 ($\downarrow$ 5)	64 ($\downarrow$ 14)
	8	4.50	40 ($\downarrow$ 1)	48 ($\downarrow$ 1)	32 ($\downarrow$ 1)	3	14	6	54 ($\downarrow$ 10)	44 ($\downarrow$ 6)	64 ($\downarrow$ 14)
	9	4.48	40 ($\downarrow$ 1)	47 ($\downarrow$ 2)	32 ($\downarrow$ 1)	2	15	2	53 ($\downarrow$ 11)	44 ($\downarrow$ 6)	61 ($\downarrow$ 17)
	10	4.45	37 ($\downarrow$ 4)	46 ($\downarrow$ 3)	28 ($\downarrow$ 5)	4	15	2	55 ($\downarrow$ 9)	44 ($\downarrow$ 6)	65 ($\downarrow$ 13)
Curriculum Learning	1	8.54	43 ($\uparrow$ 2)	52 ($\uparrow$ 3)	35 ($\uparrow$ 2)	–	–	–	54 ($\downarrow$ 10)	46 ($\downarrow$ 4)	63 ($\downarrow$ 15)
	2	8.26	43 ($\uparrow$ 2)	52 ($\uparrow$ 3)	35 ($\uparrow$ 2)	5	21	0	57 ($\downarrow$ 7)	45 ($\downarrow$ 5)	68 ($\downarrow$ 10)
	3	8.05	44 ($\uparrow$ 3)	53 ($\uparrow$ 4)	35 ($\uparrow$ 2)	4	22	4	58 ($\downarrow$ 6)	46 ($\downarrow$ 4)	70 ($\downarrow$ 8)
	4	7.97	43 ($\uparrow$ 2)	53 ($\uparrow$ 4)	34 ($\uparrow$ 1)	4	20	6	57 ($\downarrow$ 7)	46 ($\downarrow$ 4)	69 ($\downarrow$ 9)
	5	7.81	42 ($\uparrow$ 1)	52 ($\uparrow$ 3)	33 (= 0)	3	18	6	59 ($\downarrow$ 5)	47 ($\downarrow$ 3)	71 ($\downarrow$ 7)
	6	7.79	39 ($\downarrow$ 2)	48 ($\downarrow$ 1)	31 ($\downarrow$ 2)	1	15	6	57 ($\downarrow$ 7)	46 ($\downarrow$ 4)	67 ($\downarrow$ 11)
	7	7.70	39 ($\downarrow$ 2)	47 ($\downarrow$ 2)	30 ($\downarrow$ 3)	3	15	1	56 ($\downarrow$ 8)	46 ($\downarrow$ 4)	65 ($\downarrow$ 13)
	8	7.68	38 ($\downarrow$ 3)	46 ($\downarrow$ 3)	31 ($\downarrow$ 2)	1	16	2	56 ($\downarrow$ 8)	46 ($\downarrow$ 4)	66 ($\downarrow$ 12)
	9	7.63	39 ($\downarrow$ 2)	46 ($\downarrow$ 3)	32 ($\downarrow$ 1)	5	13	4	55 ($\downarrow$ 9)	44 ($\downarrow$ 6)	66 ($\downarrow$ 12)
	10	7.61	36 ($\downarrow$ 5)	44 ($\downarrow$ 5)	28 ($\downarrow$ 5)	1	14	4	55 ($\downarrow$ 9)	45 ($\downarrow$ 5)	65 ($\downarrow$ 13)

Table 3: Early-epoch learning dynamics for OLMo across different CPT strategies. We report average entity performance (overall, high-frequency, low-frequency), entities acquired, retained, forgotten, and out-of-domain changes in Avg-OOD, MMLU, and Social-IQA.